

Week 1 Day 3:

Data cleaning And Prepration

Initial Row Count: 50 records

Final Clean Row Count: 49 records

Executive Summary

The dataset underwent a data quality audit and standardization process to resolve formatting inconsistencies, text issues, missing values, duplicate records, and non-standardized date types. The clean dataset is fully prepared for aggregation, pivot tables, and statistical modeling.

Key Cleaning Operations Performed

1. Text Standardization (*Customer Name & Region*)

- Issue: Extra whitespace, inconsistent capitalization, and special characters (e.g., Ali - Khan, Ahmed Raza###, lahore, ISLAMABAD).
- Solution: Applied nested text functions to clean, standard-case, and strip non-alphanumeric artifacts:

```
=TRIM(PROPER(SUBSTITUTE(SUBSTITUTE(CLEAN(B2), "#", ""), "- ", " ")))
```

- Result: Uniform casing (Karachi, Lahore, Islamabad, Ali Khan, Ahmed Raza) with all trailing spaces removed.

2. Numeric Formatting (*Unit Price & Units*)

- Issue: Unit Price values contained text-formatted commas (e.g., 80,000, 4,500), preventing mathematical operations. Missing values (NaN/blanks) were present in the Units column.
- Solution:
- Stripped commas and converted text strings into numeric values:

```
=VALUE(SUBSTITUTE(G2, ",", ""))
```

- Imputed missing Units entries with default values (1) to prevent calculation breaks in revenue formulas.

3. Date Parsing & Standardization (Order Date)

- Issue: Mixed date types (e.g., 01/09/2026, 02-Sep-2026, September 4, 2026) stored as a combination of text strings and serial numbers.
- Solution: Used IFERROR alongside DATEVALUE to convert text representations into true serial dates while retaining valid numeric date values:

=IFERROR(DATEVALUE(E2), E2)

- Result: Standardized all date records under the uniform DD-MMM-YYYY format.

4. Deduplication & Data Integrity

- Issue: Duplicate entry detected at row 50 (ORD049).
- Solution: Applied Data -> Remove Duplicates on Order ID, resulting in 49 unique, verified customer order records.